Lopcombe Corner — HadUK-Grid Rainfall: Validation, Cycles and Trend

Rainfall: Met Office HadUK-Grid v1.3.2.ceda (release v20260512), 1 km monthly, 1963–2025, cell centred E 425500 / N 135500 containing borehole E14150 (E 425007 / N 135427) · Levels: manual-dip annual maxima 1964–1999 and corrected logger winter peaks 2001–2025 (winter 1999/2000 omitted) · analysis 20 Oct 2026

Bottom line. Over 62 winters there is **no regular cycle** in rainfall or water levels. Oct–Mar rainfall explains winter peaks well (r = 0.82 over 61 winters), every winter that reached the 97 m Pitton flood threshold had at least about 670 mm of Oct–Mar rain, and winter rainfall has risen by about 1.4 mm a year — a trend that falls just short of statistical significance.

1. Rainfall source and validation

- The single 1 km cell matches the mean of its 3×3 neighbourhood to within 1.25 mm a month, so the exact cell choice does not matter. Mean annual rainfall at the cell is 844 mm (range 548–1155 mm).
- Against the Broughton gauge (E13580, about 10 km away): Oct–Dec totals agree with r = 0.96 and within 2% overall.
- Against Tim Vaughan's Broughton Oct–Jan figures, HadUK-Grid is within −1% to +8% (table below), so the borehole cell is slightly wetter than Broughton.
- The earlier API-summed Broughton Oct–Jan totals were flawed — in 2001, 2006, 2022 and 2024 they were smaller than the Oct–Dec totals, which is impossible. HadUK-Grid replaces them.

Winter	Tim (Broughton) Oct–Jan mm	HadUK-Grid Oct–Jan mm	Difference
2000/01	534	578	+8%
2002/03	557	603	+8%
2013/14	605	634	+5%
2019/20	484	496	+2%
2023/24	509	505	−1%

2. Rainfall vs winter peak level

- Oct–Mar rainfall vs winter peak: **r = 0.82** over 61 winters; manual-dip era 1964–1999 r = 0.75; logger era 2001–2025 **r = 0.89** (better than 0.86 using Broughton).
- Fit: peak ≈ 0.039 × rain + 68.8, i.e. about +3.9 m of peak per +100 mm of Oct–Mar rain.
- The dip-era correlation is lower partly because sparse manual dips can miss the true peak.

97 m Pitton threshold. Six winters reached it, all with Oct–Mar ≥ about 670 mm. Of the seven winters above that amount, six breached; the exception was 2022/23. Five of the six breaches have come since 2000. 1976/77 (669 mm, 89.1 m) just missed the band, after the 1976 drought left the aquifer very low.

Winter	Oct–Mar rain (mm)	Peak (mAOD)	Reached 97 m?
2013/14	860	102.01	Yes
2000/01	819	101.39	Yes
2023/24	796	97.54	Yes
2019/20	716	97.25	Yes
2022/23	**709**	**92.62**	**No**
1994/95	692	98.26	Yes
2002/03	671	100.79	Yes

3. Cyclicity

Same tests as the September cyclicity check: Lomb–Scargle periodogram over 2–30 years, significance from 5,000 Monte Carlo runs against white noise (shuffled data) and red noise (fitted AR(1)), autocorrelation to lag 15, a split-half stability check and a runs test about the median.

Series	n	Strongest period	p (white)	p (red)	Halves	Runs (obs/exp)
Oct-Mar rainfall	62	3.3 yr	0.43	0.44	2.7 / 3.2 yr	32 / 32
Calendar-year rainfall	63	6.6 yr	0.74	0.75	2.3 / 7.4 yr	29 / 32
Winter peak level	61	6.7 yr	0.51	0.50	2.7 / 6.3 yr	34 / 30

- **No significant cycle.** No periodogram peak comes close to the 95% noise threshold (p ≈ 0.4–0.75).
- The water level's ~6–7 year tendency has no counterpart in Oct–Mar rainfall, whose strongest period is 3.3 years, and neither series keeps the same dominant period in its first and second halves.
- Rainfall autocorrelation stays inside the ±0.25 noise band at every lag; levels exceed it only at lag 9 (–0.26), which is about what chance gives across 15 lags. Runs tests show wet and dry winters alternating at random.
- High-rainfall winters (top 20%): 1976/77, 1989/90, 1993/94, 1994/95, 2000/01, 2002/03, 2006/07, 2009/10, 2012/13, 2013/14, 2019/20, 2022/23, 2023/24.

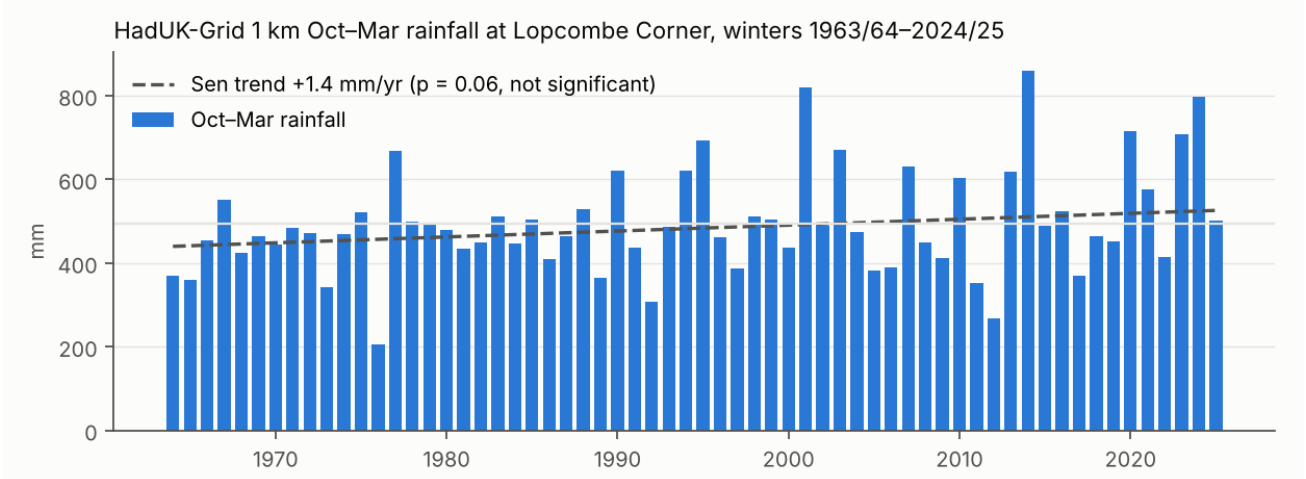
4. Trend

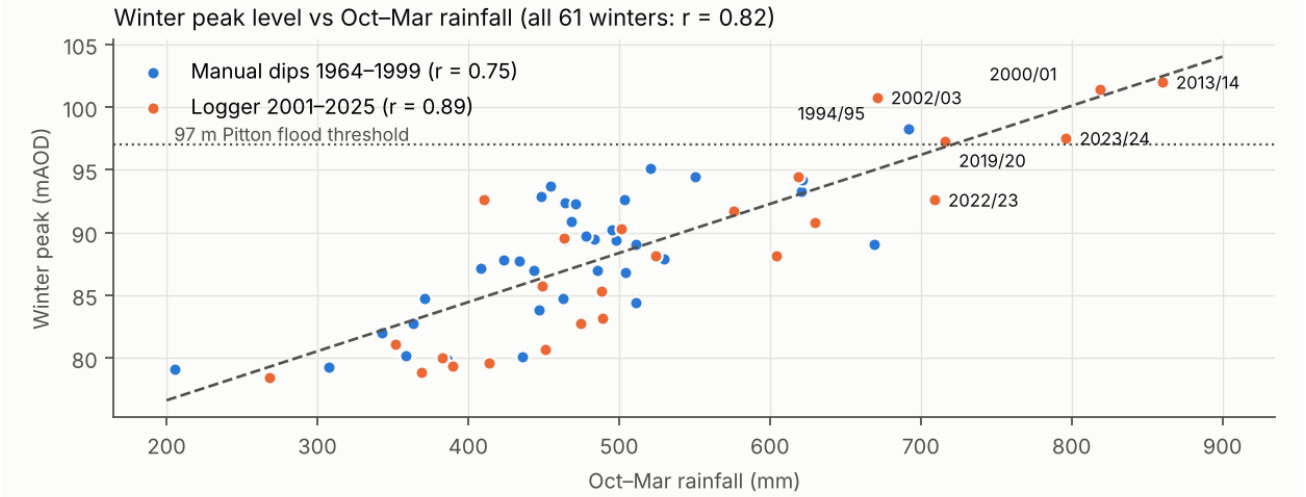
- Oct–Mar rainfall: Sen slope **+1.4 mm/yr** (about +87 mm over 62 winters, roughly 18% of the mean), Mann–Kendall p = 0.057 — suggestive but not significant at the 5% level. Eight of the 13 wettest-quintile winters fall in 2000/01 or later.
- Calendar-year rainfall: +1.7 mm/yr, p = 0.12.
- Winter peak level: no trend (Sen -0.01 m/yr, p = 0.77). Dip-era peaks may under-read where dips were sparse, which would mute any level trend.

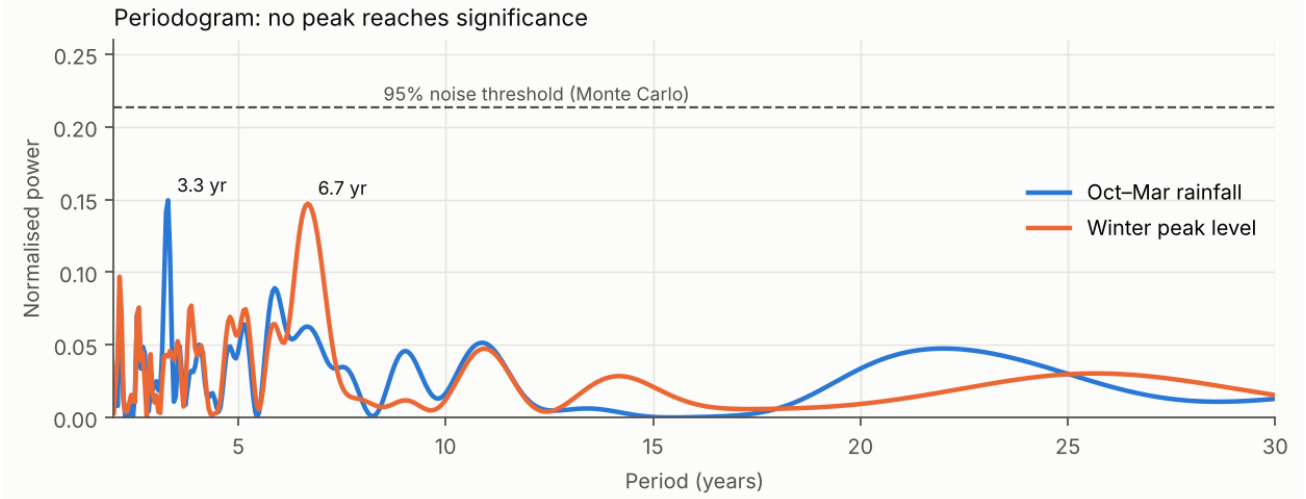
5. Files and how to reproduce

File (repo-relative)	What it is
scripts/get_haduk_1km.sh	Downloads the 63 NetCDF files from CEDA (needs a free CEDA account)
scripts/haduk_extract_cell.py	Extracts the borehole cell from the NetCDF files to CSV
scripts/haduk_rainfall_analysis.py	Rainfall vs peak, cycles, trend and chart
data/haduk/lopcombe_haduk1km_monthly_rainfall.csv	Monthly rainfall, cell and 3×3 mean, 1963–2025
output/haduk/lopcombe_winter_rain_vs_peak.csv	Winter totals and peak level per winter
output/haduk/haduk_results.json	All statistics
output/haduk/Lopcombe_HadUK_rainfall_cyclicity.png	The chart on the next page

The 2.6 GB of NetCDF files are kept on the external SSD (/Volumes/Crucial X9 Pro for Mac/HadUK-Grid), not in the repo. A possible next step is adding the start-of-winter groundwater level to the 62-winter model, which could explain wet winters such as 2022/23 that stayed below the threshold.


HadUK-Grid 1 km Oct–Mar rainfall at Lopcombe Corner, winters 1963/64–2024/25
Sen trend +1.4 mm/yr (p = 0.06, not significant)
Oct–Mar rainfall
mm
0
200
400
600
800
1970
1980
1990
2000
2010
2020


Winter peak level vs Oct–Mar rainfall (all 61 winters: r = 0.82)
Manual dips 1964–1999 (r = 0.75)
Logger 2001–2025 (r = 0.89)
97 m Pitton flood threshold
1994/95
2002/03
2000/01
2013/14
2023/24
2019/20
2022/23
Winter peak (mAOD)
Oct–Mar rainfall (mm)
80
85
90
95
100
105
200
300
400
500
600
700
800
900


Periodogram: no peak reaches significance
95% noise threshold (Monte Carlo)
3.3 yr
6.7 yr
Oct–Mar rainfall
Winter peak level
Normalised power
Period (years)
0.00
0.05
0.10
0.15
0.20
0.25
5
10
15
20
25
30